

Informatics 141

Computer Science 121

Information Retrieval

Lecture 2

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These course materials borrow, with permission, from those of Prof. Cristina Videira Lopes, Prof. Alberto Krone-Martins, Addison Wesley 2008, Chris Manning, Pandu Nayak, Hinrich Schütze, Heike Adel, Sascha Rothe, Jerome H. Friedman, Robert Tibshirani, and Trevor Hastie. Powerpoint theme by Prof. André van der Hoek.

Announcements

- Assignment 1 has been posted.

- Introduction
- Information Retrieval and Web Search

Our definition of information retrieval

The activity of searching and extracting information from a collection of information resources.

In this class : textual information from text documents

Web Search Basics

Introduction to Information Retrieval

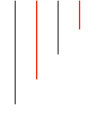
Web Search Engines

- How often do you use web search engines?

Web Search Engines

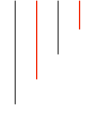
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Web Search Engines



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- The Web is both a technology artifact and a social environment
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or even Internet = Google search
- Search engines make aggregation of interest possible:
 - Create incentives for niche players

- Search interaction makes “unlimited selection” stores possible
 - Amazon, Netflix, etc.
- Pew Research Center on *Search Engine Use 2012*

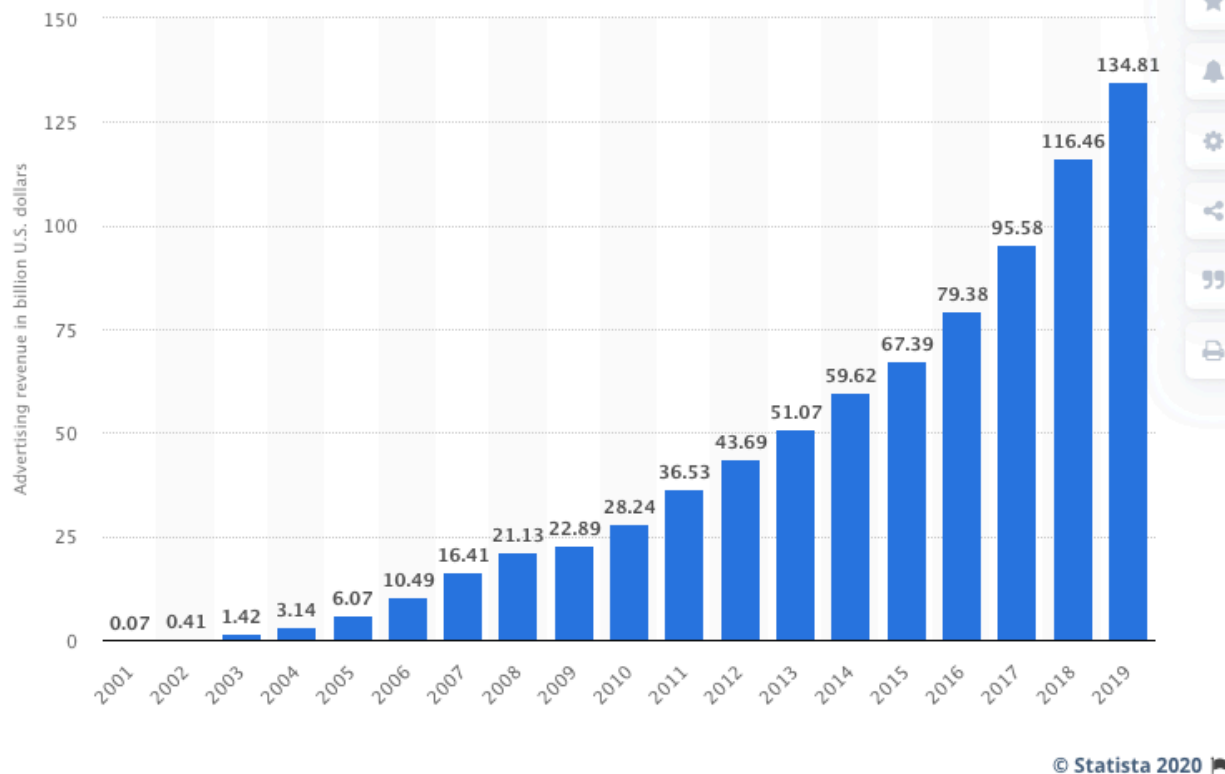
*Search engines remain popular—and users are more satisfied than ever with the quality of search results—but many are **anxious about the collection of personal information by search engines and other websites.***

Web Search Engines

- Search turned out to be the [first] best mechanism for advertising on the Web:
 - web advertising revenue: Google (2019, global) ??????????

Web Search Engines

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Web Search Engines

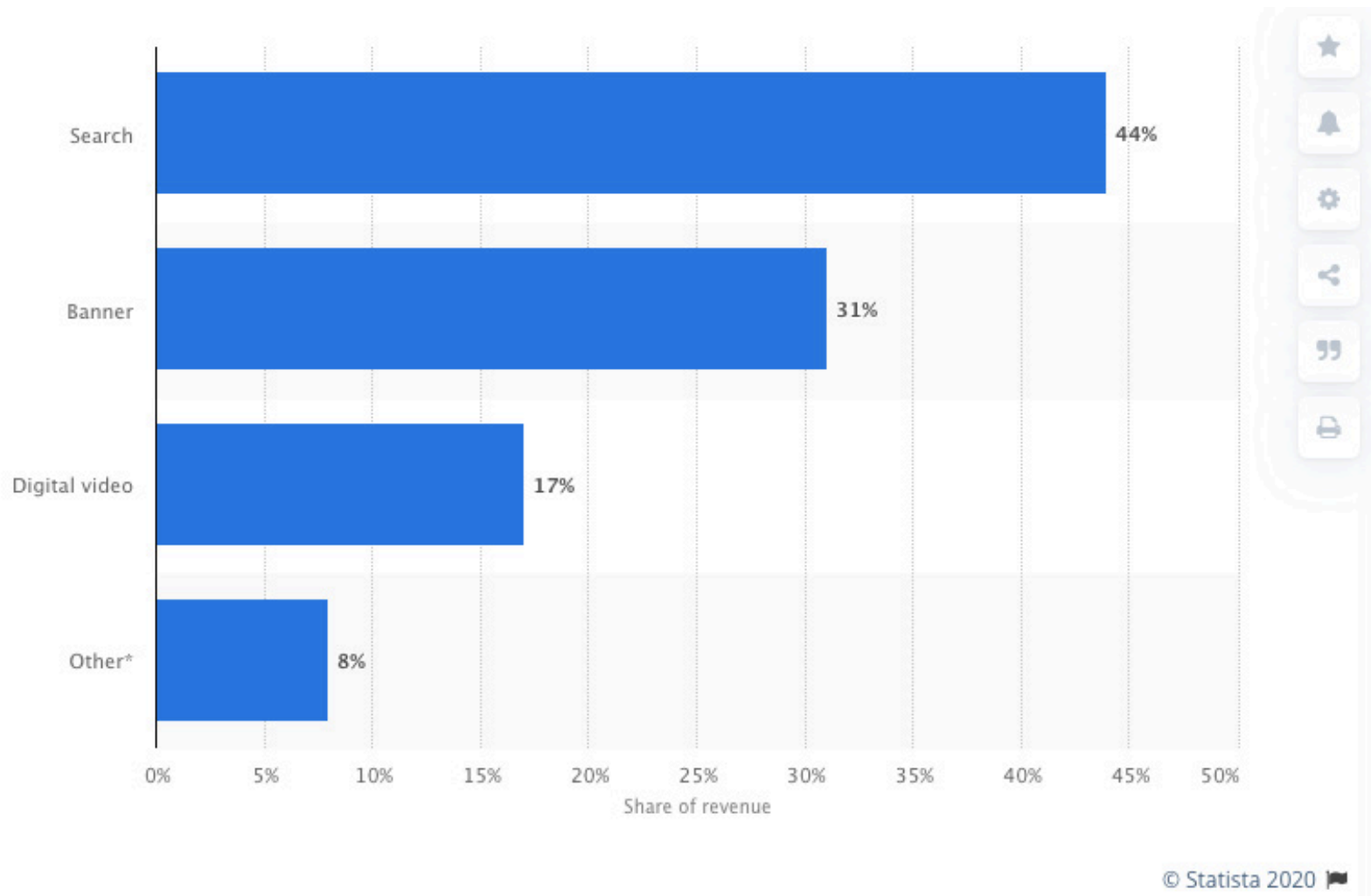
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 - Business model for ads changed over time
- Statistics and facts on the Digital Advertising Industry in the U.S. <https://www.statista.com/topics/1176/online-advertising/> ::
 - *By 2021, digital ad. is projected to add up to more than **330 billion** U.S. dollars worldwide*

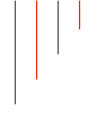
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Web Search Engines

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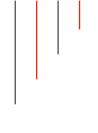




- Web search and Information retrieval
 - Essential to modern life
 - Central for your future professional life in Industry and Academia

Information Retrieval and Web Search

Introduction to Information Retrieval



- Information retrieval is a **field** concerned with the structure, analysis, organization, storage, searching, and retrieval of information (Salton, 1968)
- Primary focus: text
 - Scholarly papers, books, news, email, Web pages
- Other media:
 - Audio, images, video



- **Search (this course)**
- Filtering or tracking
 - Detecting documents of interest
- Classification
 - Labeling documents with pre-existing labels
- Question answering
 - Beyond searching for documents, searching for answers

Types of Search Engines

- **Web search (this course)**
- Vertical search
- Enterprise search
- Desktop search
- Peer-to-peer search

Database Engines vs. Search Engines

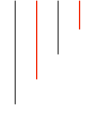
- What is the difference between a database search and a search engine?

Database Engines vs. Search Engines

- “Find accounts whose balance is greater than \$100”
 - Requires data to be **well-structured** and **well-defined**
 - Typically involves not just **text** but **numbers**
 - Requires use of **formal language and logic**
- “Canvas UCI” or
“Find how to merge two lists in Python”
“Find the graph of $y=\log(x)$ ”
 - **Vague** and varied information needs
 - **Informal** and **ad-hoc queries**
 - Requires engine to **infer meaning** of words and sentences

Core Issue: Relevance

- Textual similarity **[databases stop here !!]**
- Document context
 - Origin
 - Author
 - Popularity
 - ...
- User context
 - Geographic location
 - Prior queries
 - Age
 - Preferred language
 - ...
- Query context
 - Special symbols



- Conceptual designs of what to pay attention to when matching a query and a document
 - E.g. grammar vs. raw text, what parts of context, etc.
- Good retrieval model: finds documents relevant to the person who submitted the query
- Basis for *ranking algorithms*

Classic IR Assumptions

- **Corpus:** fixed document collection
- Goal: retrieve information content relevant to the information need

- “Relevance”
 - For each **query Q**, and stored **document D**, **there exists** a relevance **score $R(Q, D)$**
 - **Maximize $R(Q, D)$**
 - Context is ignored
 - User is ignored
 - Corpus is static and centralized



- The Web is huge (**cannot store** in any centralized memory!)
 - Corpus is not centralized!
- The Web changes all the time (**needs to update** constantly!)
 - Corpus is not static!
- There is information to avoid (adversarial IR!)
 - Context cannot be ignored!
- One interface for hugely divergent needs :
 - User cannot be ignored!